

**SIMATS ENGINEERING**  
**Department of Computer Science & Engineering**  
**ASSESSMENT TOOL 2- COMPARATIVE ANALYSIS**

|                |                                                                                                                                                                                                                                                    |               |                                           |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------------|
| Course Code:   | CSA12                                                                                                                                                                                                                                              | Course Title: | Computer Architecture                     |
| CO Assessed:   | CO4                                                                                                                                                                                                                                                | Assessment:   | ASSESSMENT TOOL2-<br>COMPARATIVE ANALYSIS |
| Weightage:     | 30%                                                                                                                                                                                                                                                | Total Marks:  | 25                                        |
| CO4 Statement: | Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics. |               |                                           |
| Student Name:  | VaAdarsh Redd                                                                                                                                                                                                                                      | Reg.No.:      | 192525412                                 |
| Department:    | B.Tech FI/ML                                                                                                                                                                                                                                       | Date:         | 24/08/26                                  |

**ANNEXURE A**  
**COMPARATIVE ANALYSIS**

1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.
2. Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.
3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.
4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.
5. Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

**Code of Conduct:**

I VaAdarsh Redd (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.

V. Adaksh Reddy  
Signature of the student:

## MARKS ALLOCATION

|    | Scope of Items<br>(20%) | Comparison<br>Depth (25%) | Use of<br>Evidence<br>(20%) | Critical Thinking<br>(20%) | Clarity of<br>Presentation (15%) |
|----|-------------------------|---------------------------|-----------------------------|----------------------------|----------------------------------|
| Q1 | 0.8                     | 1                         | 1                           | 1                          | 0.75                             |
| Q2 | 0.8                     | 1                         | 1                           | 1                          | 0.75                             |
| Q3 | 1                       | 1.25                      | 0.8                         | 1                          | 0.25                             |
| Q4 | 1                       | 1.25                      | 0.8                         | 1                          | 0.5                              |
| Q5 | 0.5                     | 1                         | 1                           | 1                          | 0.5                              |

20.75

## RUBRICS FOR CALCULATION:

| Criteria         | Weightage | Level 4 – Exemplary                                                    | Level 3 – Proficient                          | Level 2 – Developing                       | Level 1 – Beginning               |
|------------------|-----------|------------------------------------------------------------------------|-----------------------------------------------|--------------------------------------------|-----------------------------------|
| Scope of Items   | 20%       | Selects highly relevant items with clear scope and justification       | Selects appropriate items with minor gaps     | Selection is limited or partially relevant | Items are unclear or not relevant |
| Comparison Depth | 25%       | Provides detailed and comprehensive comparison across multiple aspects | Provides adequate comparison with some depth  | Limited comparison with few aspects        | Very minimal or no comparison     |
| Use of Evidence  | 20%       | Supports comparison with accurate data, examples, or references        | Uses some supporting evidence with minor gaps | Limited or weak supporting evidence        | No supporting evidence provided   |